

Optical Intelligent Reflecting Surface-Aided Visible Light Positioning based on RSS Fingerprint

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Abstract—In this paper, an optical intelligent reflecting surface (OIRS)-aided visible light positioning (VLP) system that adopt received signal strength (RSS) fingerprint method is proposed to enhance the positioning accuracy. We employ OIRS to improve the minimal RSS difference between neighboring positions in conventional RSS fingerprint-based VLP systems, involving a two-stage positioning scheme where the OIRS is strategically aligned to refine the positioning accuracy. Subsequently, the considered problem is formulated to maximize the minimum Euclidean distance by optimizing the OIRS configuration, which is then solved by an successive convex approximation (SCA)-based algorithm. Moreover, simulation results demonstrate that the proposed OIRS-assisted VLP system substantially outperforms conventional benchmarks, thereby achieving superior positioning accuracy.

Index Terms—Visible light positioning, optical intelligent reflecting surface, received signal strength, minimum Euclidean distance, successive convex approximation.

I. INTRODUCTION

Visible light positioning (VLP) has garnered significant attention due to its high positioning accuracy and broad application prospects [1]. Since optical signal owns shorter wavelength, VLP offers higher positioning precision, surpassing WiFi or Bluetooth [2]. Besides, traditional radio frequency-based technologies suffer from multipath effects and environmental interference, while visible light signals significantly reduce the impact, thereby enhancing the positioning accuracy [3]. Moreover, the widespread deployment of light emitting diode (LED) in indoor scenarios facilitates the utilization of existing lighting infrastructure for VLP, which saves both hardware costs and deployment efforts [4]. Consequently, VLP features inherent advantages for indoor positioning.

Received signal strength (RSS)-based method is a fundamental algorithm in VLP [3], which includes several branch algorithms, such as trilateration, proximity, and fingerprint [1]. Specifically, the trilateration method obtains the target position by calculating the distance from each LED, while deriving distance directly from RSS is complex in VLP [5]. Besides, the proximity approach selects the position of the LED with the largest RSS as the result, performing poor accuracy [6]. Notably, the fingerprint scheme collects RSS values at predefined locations and determines the position by comparing them with the measured one, providing more practical than others in VLP due to its simplicity of implementation [7].

Recently, intelligent reflecting surface (IRS) is an emerging technology capable of dynamically reconfigure wireless channels and has substantially motivated academia to engage in the research [8]. Furthermore, the optical IRS (OIRS) extends the spectrum to the optical band by dynamically adjusting the reflection characteristics of optical signals to optimize the optical communication performance [9]. Generally, OIRS comprises multiple programmable reflecting elements, which can modify the reflection angles and the signal intensities through electronic or mechanical means [10]. Moreover, by strategically adjusting reflection characteristics and reconfiguring optical channels, OIRS can enhance the signal coverage and improve the signal strength [11].

Besides, the potential of OIRS to revolutionize VLP systems has also sparked considerable research interest [12]. Beyond the inherent characteristics of VLP systems, the implementation of OIRS can achieve even higher precision. However, integrating VLP and OIRS is an emerging field with limited research. Specifically, literature [13] primarily utilized OIRS to address positioning issues under occlusion conditions, proposing a sparse Bayesian learning (SBL)-based algorithm to jointly optimize position estimation, noise variance, and sparsity. Besides, the Cramér-Rao lower bound (CRLB) expression and maximum likelihood (ML) estimator for general 3D positioning in the presence of arbitrarily configured OIRS were derived in [14], and an optimal adjustment approach to maximize the received power of OIRS was proposed.

This study explores an integration strategy of OIRS and VLP adopting RSS fingerprint method for the first time to the best of our knowledge. Firstly, OIRS is leveraged to alter the RSS values at certain positions to increase the difference of RSS fingerprints between adjacent locations, thus reducing mispositioning. Secondly, a two-stage positioning scheme is proposed to fully exploit OIRS while decreasing the computational complexity. Moreover, the OIRS alignment problem is formulated to maximize the evaluation indicator, which is determined by the minimum Euclidean distances of RSS fingerprints. Thirdly, an successive convex approximation (SCA)-based algorithm is proposed to solve the problem. Furthermore, numerical results prove the validity of the proposed algorithm and indicate the significant improvement in the positioning accuracy attributed by OIRS.

The remainder of this paper is organized as follows. Sec-

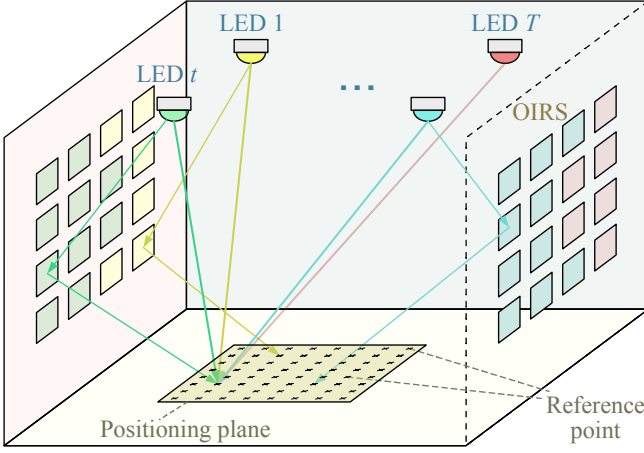


Fig. 1. The architecture of the OIRS-aided VLP system.

tion II describes the system model. Section III outlines the RSS fingerprint and details the proposed two-stage scheme. Section IV introduces the evaluation indicator, formulates the problem, and then presents the SCA-based optimization algorithm. Section V conducts various simulations and proves the improvement of the positioning accuracy by OIRS. Finally, Section VI concludes the paper.

II. SYSTEM MODEL

As illustrated in Fig. 1, T LEDs are deployed on the ceiling at known positions, while two OIRSs consisting of N elements is fixed on opposite walls. Besides, the user equipped with the photodetector (PD) is possibly located on a positioning plane with R pre-determined locations, called the *reference points*, which are utilized in the proposed VLP system. Then, the sets $\mathcal{T}$, $\mathcal{N}$, and $\mathcal{R}$ are defined to denote the index of LEDs, OIRS units, and reference points, respectively.

In this paper, LEDs adopt the electrical signals of distinct frequencies as the positioning signals and modulate them into optical signals for transmission. Besides, the PD receives the optical signal and converts it back to the electrical signal, which can be separated by frequency utilizing signal processing or filtering techniques. Moreover, due to the non-negative nature of the optical signals, a direct current (DC) component should be added. Thus, the transmitted signals can be represented as

$$\mathbf{x} = \mathbf{s} + \mathbf{b}, \quad (1)$$

where each element of $\mathbf{s} \in \mathbb{R}^{T \times 1}$ denotes the signals with normalized amplitude at different frequencies. Moreover, $\mathbf{b}$ is the DC for the signals of all LEDs, with all elements being identical to ensure the uniform illumination.

Without loss of generality, the line-of-sight (LoS) channel gain from the t -th LED to the r -th reference point typically conform to the Lambertian radiant model [9], denoted as $h_{r,t}$. Besides, the channel vector of the r -th reference point is defined as $\mathbf{h}_r = [h_{r,1}, h_{r,2}, \dots, h_{r,T}] \in \mathbb{R}_+^{1 \times T}$, $r \in \mathcal{R}$.

Therefore, the received signal of the LoS path from different LEDs at the r -th reference point can be expressed as

$$\mathbf{y}_r^{(\text{LoS})} = \mathbf{h}_r \mathbf{x}. \quad (2)$$

In this study, a binary matrix $\mathbf{A} \triangleq [a_{r,n,t}] \in \mathbb{B}^{R \times N \times T}$ is introduced to indicate the alignment between the OIRS units and the transceivers [11]. Specifically, $a_{r,n,t} = 1$ suggests the assignment of the n -th OIRS unit to the t -th LED and the r -th reference point. Given that each OIRS unit can align with only one LED and one reference point, the constraint $\sum_{r=1}^R \sum_{t=1}^T a_{r,n,t} = 1, \forall n \in \mathcal{N}$ holds.

Moreover, the specular reflection utilizes an “additive” model [10], where the path length is the sum of the distances from the LED to the OIRS unit and from the OIRS unit to the reference point. Consequently, the OIRS-reflected channel gain between the t -th LED and the r -th reference point via the n -th OIRS units is denoted as $g_{r,n,t}$. To simplify the expression, the alignment relationship is incorporated into the channel gain. Explicitly, a composite matrix $\mathbf{G}(\mathbf{A}) \triangleq [\sum_{n=1}^N a_{r,n,t} g_{r,n,t}] \in \mathbb{R}_+^{R \times T}$ is defined, and $\mathbf{g}_r(\mathbf{A})$ is the r -th row of $\mathbf{G}(\mathbf{A})$. Therefore, the received signal of the r -th reference point via OIRS-reflected path is given by

$$\mathbf{y}_r^{(\text{Ref})} = \mathbf{g}_r(\mathbf{A}) \mathbf{x}. \quad (3)$$

III. PROPOSED RSS FINGERPRINT-BASED SCHEME

A. RSS Fingerprint for the OIRS-aided VLP

The positioning scheme utilized in this study employs the RSS fingerprint method. Generally, the RSS vector is defined to represent the received RSS values from different LEDs after filtering. Moreover, fingerprint-based VLP systems consist of two processes: the offline survey and the online positioning. The former collects and records the RSS vectors at the reference points to construct the database. The RSS vector stored in the database is known as the *RSS fingerprint*. The online positioning involves the measurement of the RSS vector at the current location, which is then compared to the fingerprint vectors to find the closest, thereby determining the position. Typically, the process includes calculating the Euclidean distance among the vectors, with smaller distances indicating closer matches.

This study employs OIRS to assist positioning by increasing the differences of RSS fingerprints among the reference points. By adjusting the alignment of the OIRS with each location, the issue of similar RSS vectors at different positions can be significantly mitigated, thereby enhancing the positioning accuracy. Moreover, two OIRSs are deployed on opposite walls to ensure that all reference points receive adequate coverage and signal improvement.

After the received signal $y_r = y_r^{(\text{LoS})} + y_r^{(\text{Ref})} + v_r$ is bandpass filtered at T different frequencies of all LEDs to obtain the signal vector, where $v_r \sim \mathcal{N}(0, \sigma^2)$ denotes the additive white Gaussian noise, the strengths of the signal vector are determined by the gains of the normalized signals $\mathbf{s}$ at various baseband frequencies, while the DC component $\mathbf{b}$ is filtered out. Therefore, these strengths are defined as the

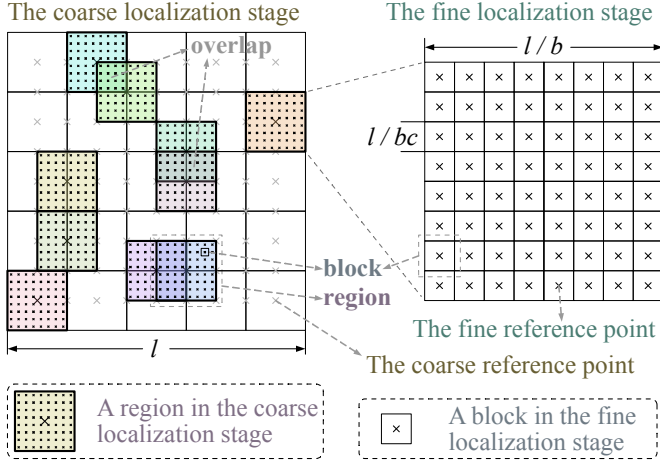


Fig. 2. Spatial partitioning of the proposed two-stage scheme.

RSS values in this paper, which are considered as the RSS fingerprint for the certain location. Consequently, the RSS fingerprint of the r -th reference point can be given by

$$\mathbf{f}_r(\mathbf{A}) = \mathbf{h}_r + \mathbf{g}_r(\mathbf{A}) \in \mathbb{R}^{1 \times T}, \quad (4)$$

with each element representing the signal strength from the corresponding LED at the r -th reference point.

B. Proposed Two-Stage Positioning Scheme

The introduction of OIRS boosts the positioning performance. However, there is limitation in effective illumination scope of an OIRS unit. Firstly, the reference point is dense for higher positioning accuracy, typically much more than the OIRS units. Thus, the improvement by OIRS is limited. Besides, the alignment of OIRS brings high computational complexity. To address these, we propose a two-stage positioning scheme consists of a coarse localization stage and a OIRS-aided fine localization stage, as shown in Fig. 2.

Initially, we define a square *region* that contains the possible locations of the user on the ground with side length l . Then, the region is divided into several *blocks* due to the impracticality of recording the RSS vector for every possible location. Moreover, the center of region is treated as the reference point in the coarse stage, while the center of block is considered as the reference point for the fine stage.

In the coarse localization stage, the most possible region is determined, with the side length being l/b , where b being an integer noticeably larger than 1. During the offline survey, the RSS fingerprints at the reference points are collected and stored. In the online positioning, by comparing the current RSS vector with those in the database, the best region is selected as candidates for the fine localization stage. Notably, the regions are arranged in an overlapping pattern to increase the system robustness. Specifically, the spacing between regions is $l/2b$ instead of l/b . This arrangement mitigates localization errors caused by incorrect region selection, as each block is contained within four regions. Even if the user is assigned to

a suboptimal region during the coarse localization stage, the correct block is still included.

The fine localization stage considers the selected region rather than the entire room as the positioning plane, which contains $R = c \times c$ blocks with the length of l/bc . Moreover, each region stores an OIRS alignment matrix and an RSS fingerprint database, with much smaller size than those in the origin RSS fingerprint-based VLP aided by OIRS, thereby reducing the computational complexity. Then, the online positioning select the most suitable block where the user is likely located. Particularly, OIRSs are only utilized during the fine localization stage instead of the coarse stage, as the regions are too large for the OIRS units to cover. Therefore, the subsequent problems only focus on the fine localization stage.

IV. PROBLEM FORMULATION AND PROPOSED ALGORITHM

A. Evaluation Indicator

In the proposed scheme, the RSS vector measured by the user should be closest to the fingerprint at the nearest reference point, resulting in accurate localization. However, due to the noise, the measured vector may be closer to a different fingerprint, leading to mispositioning. Consequently, increasing the Euclidean distance between different fingerprints can significantly enhance the positioning accuracy.

In our discussion, the Euclidean distance from the i -th fingerprint to the j -th fingerprint is considered, denoted as $e_{i,j}$. Assume that a user is at the i -th reference point, as long as the measured RSS is within half the Euclidean distance of $\min_j e_{i,j}$ from $\mathbf{f}_i$, then the correct reference point is found. Thus, the accuracy improves when $\min_j e_{i,j}$ increases. To represent the overall localization performance, we utilize the minimum $e_{i,j}$ among all $i, j \in \mathcal{R}$ as the evaluation indicator. Specifically, this method directly improves the worst $e_{i,j}$ through OIRS alignment, ensuring a lower bound for the localization performance across the entire region and preventing the scenarios of exceptionally degraded performance. Therefore, the evaluation indicator is defined by taking the square of the minimum value among all the Euclidean distances as

$$e_{\min} = \min_{i,j \in \mathcal{R}, i \neq j} e_{i,j}^2 = \min_{i,j \in \mathcal{R}, i \neq j} \|\mathbf{f}_i - \mathbf{f}_j\|_2^2. \quad (5)$$

B. Problem Formulation

This study endeavors to improve the positioning performance in the fine localization stage by optimizing the alignment matrix $\mathbf{A}$ to maximum the minimum Euclidean distance. Thus, the optimization problem is formulated as

$$(\mathbf{P}) : \max_{\mathbf{A}} e_{\min} \quad (6)$$

$$\text{s.t. } a_{r,n,t} \in \mathbb{B}, \quad \forall r \in \mathcal{R}, n \in \mathcal{N}, t \in \mathcal{T}, \quad (6a)$$

$$\sum_{r=1}^R \sum_{t=1}^T a_{r,n,t} = 1, \quad \forall n \in \mathcal{N}, \quad (6b)$$

$$\sum_{n=1}^N \sum_{t=1}^T a_{r,n,t} \leq R_0, \quad \forall r \in \mathcal{R}, \quad (6c)$$

where constraints (6a) and (6b) arise from the properties of the OIRS alignment matrix. Moreover, constraint (6c) is introduced to prevent an excessive allocation of OIRS units to a single reference point, where R_0 represents the maximum number of OIRS units that can be assigned to each reference point. This constraint is implemented to ensure fairness in the allocation of OIRS and to prevent excessive illumination that could potentially harm users' eyes, which aligns with practical application scenarios.

C. Proposed Optimization Algorithm

The OIRS alignment problem (**P**) is an integer programming problem due to the discrete nature of $\mathbf{A}$, making it challenging to solve. To facilitate the utilization of optimization algorithms, the constraint (6a) is relaxed to a continuous one, while a relaxed matrix $\tilde{\mathbf{A}}$ is defined and satisfies

$$0 \leq \tilde{a}_{r,n,t} \leq 1, \quad \forall r \in \mathcal{R}, n \in \mathcal{N}, t \in \mathcal{T}. \quad (7)$$

Consequently, the origin OIRS alignment problem (**P**) is reformulated to a relaxed problem (**P1**) as

$$(\mathbf{P1}) : \max_{\tilde{\mathbf{A}}} e_{\min} \quad (8)$$

s.t. (7),

$$\sum_{r=1}^R \sum_{t=1}^T \tilde{a}_{r,n,t} = 1, \quad \forall n \in \mathcal{N}, \quad (8a)$$

$$\sum_{n=1}^N \sum_{t=1}^T \tilde{a}_{r,n,t} \leq R_0, \quad \forall r \in \mathcal{R}. \quad (8b)$$

Despite problem (**P1**) being a continuous optimization problem, the non-convexity of the objective function still makes it challenging to solve. On the one hand, consider $e_{i,j}$, which is a quadratic function of $\tilde{\mathbf{A}}$ with a positive coefficient in the quadratic term, making $e_{i,j}$ a convex function. However, the maximization of a convex function is not in the form of a convex optimization problem. On the other hand, $e_{\min}$ can be considered as the lower bound of all $e_{i,j}$ ($\forall i, j \in \mathcal{R}$) which are convex function, making it impossible for $e_{\min}$ to be a concave function. Consequently, (**P1**) remains a non-convex optimization problem and is difficult to solve directly.

Addressing a non-convex problem, particularly a complex optimization problem, generally benefits from the SCA method, which is a recognized and practical approach [15]. The core idea of SCA is to tackle a difficult problem by solving a series of convex approximation sub-problems. Specifically, at a feasible point $\hat{\mathbf{x}}$ of the objective function $y(\mathbf{x})$, a surrogate function $\bar{y}(\mathbf{x}|\hat{\mathbf{x}})$ replaces the original function to convert the problem into a convex optimization problem. Subsequently, the new problem is easier to address optimally, and the solution $\mathbf{x}' = \arg \min_{\mathbf{x}} \bar{y}(\mathbf{x}|\hat{\mathbf{x}})$ then serves as the starting point for the next iteration. This iterative process eventually yields a suboptimal solution to the original problem.

In the SCA method, the two functions exhibit the same value at the current point. Besides, the surrogate function needs to exhibit gradient equality to the original problem. To construct

Algorithm 1 Generate discrete matrix based on probability

Input: The continuous alignment matrix $\tilde{\mathbf{A}}$, the probability threshold p_0 .

Output: Binary alignment matrix $\mathbf{A}$.

- 1: **Init:** iteration rounds $n \leftarrow 0$, set elements in $\tilde{\mathbf{A}}$ to 0 if less than p_0 , $\mathbf{A} \leftarrow \mathbf{0}$.
- 2: **repeat**
- 3: $n \leftarrow n + 1$
- 4: Calculate the sum of all elements in $\mathbf{A}$ with index n , or $p_{n,\text{sum}} = \sum_{r=1}^R \sum_{t=1}^T \tilde{a}_{r,n,t}$;
- 5: Normalize the n -th column of the matrix $\tilde{\mathbf{A}}$, $\tilde{a}_{r,n,t} \leftarrow \tilde{a}_{r,n,t}/p_{n,\text{sum}}$;
- 6: Consider the n -th column of $\tilde{\mathbf{A}}$ as probabilities, then randomly select one pair of index r, t based on the probabilities as $p(r, t) = \tilde{a}_{r,n,t}$;
- 7: Update the value $a_{r,n,t} \leftarrow 1$.
- 8: **until** $n = N$

the surrogate function, the gradient of the objective function should be derived initially as

$$(\nabla_{\tilde{\mathbf{A}}} e_{\min})_{r,n,t} = \begin{cases} 2(f_{r,t} - f_{j_m,t})g_{r,n,t}, & \text{if } r = i_m, \\ 2(f_{r,t} - f_{i_m,t})g_{r,n,t}, & \text{if } r = j_m, \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where i_m and j_m denote the Euclidean distance between the i_m -th and the j_m -th fingerprints is the smallest for the current alignment matrix $\tilde{\mathbf{A}}$, and $e_{i_m,j_m} = e_{\min}$. Since the gradient only considers e_{i_m,j_m} , it has non-zero elements only in its i_m -th and j_m -th rows, with all other values being zeros.

Based on the above discussion, the surrogate function in the proposed SCA method is defined as

$$\bar{e}_k(\tilde{\mathbf{A}}) = e_{\min}(\mathbf{A}^{(k-1)}) + \nabla_{\mathbf{A}} e_{\min}(\mathbf{A}^{(k-1)}) * (\tilde{\mathbf{A}} - \mathbf{A}^{(k-1)}) - \frac{1}{2\alpha} \|\tilde{\mathbf{A}} - \mathbf{A}^{(k-1)}\|_2^2, \quad (10)$$

where k is the iteration number, and $\mathbf{A}^{(k-1)}$ is the solution obtained from the previous iteration. Therefore, the final surrogate problem can be expressed as

$$(\mathbf{P2}) : \max_{\tilde{\mathbf{A}}} \bar{e}_k(\tilde{\mathbf{A}}) \quad (11)$$

s.t. (7), (8a), (8b).

However, merely using the SCA method does not entirely resolve the issue, because the continuous $\tilde{\mathbf{A}}$ must be converted into a discrete matrix that meets constraint (6a). Considering the characteristics of this problem, this study proposes a method to generate a discrete matrix based on probability. Specifically, the elements of the continuous matrix $\tilde{\mathbf{A}}$ are treated as probabilities, that is, $p(a_{r,n,t} = 1) = \tilde{a}_{r,n,t}$, with $\sum_{r=1}^R \sum_{t=1}^T p(a_{r,n,t} = 1) = 1$. After obtaining $\tilde{\mathbf{A}}$, for each n , only one $a_{r,n,t}$ can be 1, with its probability equal to $\tilde{a}_{r,n,t}$ in $\tilde{\mathbf{A}}$. Furthermore, a probability threshold is introduced, detailed in **Algorithm 1**. Thus, by adopting this method, a required discrete matrix can be probabilistically generated.

Algorithm 2 SCA-based OIRS Alignment Optimization**Input:** CSI matrices $\mathbf{H}$ and $\mathbf{G}$, execution times L .**Output:** $\mathbf{A}$.

- 1: **Init:** iteration rounds $k \leftarrow 0$, the convergence threshold ϵ , $\mathbf{A}^{(0)}$ is feasible solution to $(\mathbf{P})$.
- 2: **repeat**
- 3: $k \leftarrow k + 1$
- 4: Create the surrogate function $\bar{e}_k(\tilde{\mathbf{A}})$ base on (10);
- 5: Formulate the surrogate problem utilizing $\bar{e}_k(\mathbf{A})$ and solve it as a quadratic convex optimization problem to obtain the optimal solution $\tilde{\mathbf{A}}$.
- 6: Generate discrete matrix based on probability $\tilde{\mathbf{A}}$ for L times to obtain L matrices $\mathbf{A}$, then select the $\mathbf{A}$ with the highest $e_{\min}(\mathbf{A})$ as $\mathbf{A}^{(k)}$.
- 7: **until** $|e_{\min}(\mathbf{A}^{(k)}) - e_{\min}(\mathbf{A}^{(k-1)})| \leq \epsilon$
- 8: $\mathbf{A} \leftarrow \mathbf{A}^{(k)}$.

TABLE I
SIMULATION PARAMETERS

Parameters	Values
The Lambertian index, m	1
The responsivity of PD, ρ	1
The reflectivity of OIRS, δ	0.95
The optical filter gain, g_f	1
The area of PD, S	1 cm ²
The semi-angle of FoV, Φ_{FoV}	80°
The refractive index of PD, η	1.5
The max units per reference point, R_0	6
The probability threshold, p_0	0.02

Following the above discussion, the overall OIRS alignment optimization approach utilizing the SCA-based method is illustrated in **Algorithm 2**, where in each iteration, the surrogate optimization objective is given based on the current $\mathbf{A}$ in step 4. Subsequently, the result in step 5 is adopted to probabilistically generate a discrete matrix in step 6.

V. NUMEROUS RESULTS

This section presents simulation results by MATLAB to validate the feasibility of the proposed algorithm and demonstrate the impact of OIRS on enhancing the localization performance in the proposed VLP system. The simulation assumes a room size of 4 m × 4 m × 3 m, where the localization system employs four LEDs, uniformly distributed within the room at coordinates (1 m, 1 m, 3 m), (1 m, 3 m, 3 m), (3 m, 1 m, 3 m), and (3 m, 3 m, 3 m). Additionally, OIRS units are centrally placed on the walls at $x = 0$ m and $x = 4$ m. Each OIRS unit measures 10 cm × 10 cm, with a 10 cm gap between adjacent units, forming a near-square array on each wall. Moreover, given that each OIRS unit has a side length of 10 cm, the floor is divided into blocks of 10 cm × 10 cm, resulting in a total of 1,600 blocks. Besides, b and c are set to 5 and 8, respectively, so there are $(2b - 1)^2 = 81$ regions in total, and each region covers an area of 0.8 m × 0.8 m, consisting of 8 × 8 blocks. The regions are represented by a pair of index from 1 to 8, while regions (0, 0), (0, 8), (8, 0), and (8, 8)

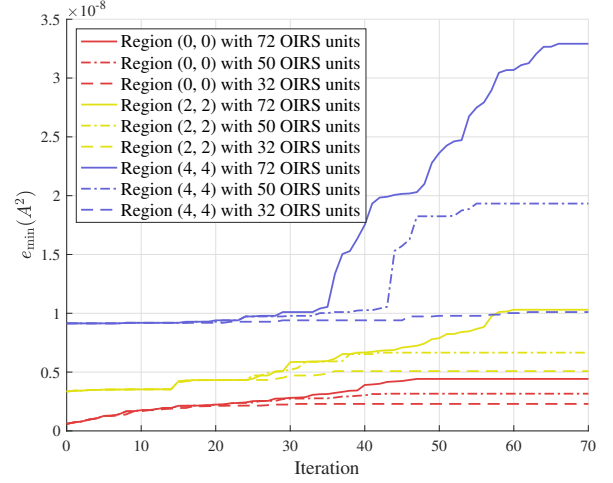


Fig. 3. The convergence of the proposed SCA-based algorithm.

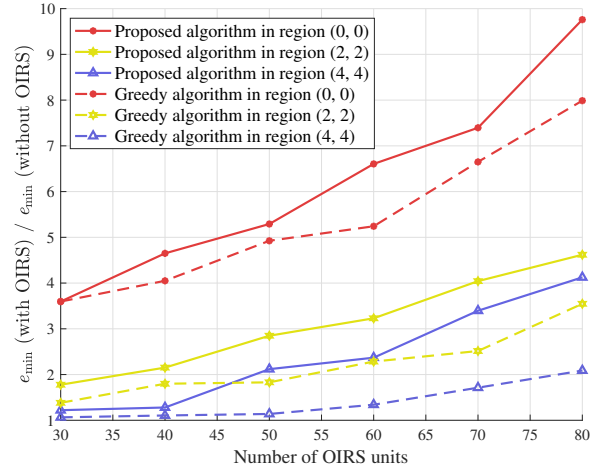


Fig. 4. The multiplication factor with various numbers of OIRS units.

are the corners of the room. Moreover, additional simulation parameters are summarized in Table I.

Fig. 3 depicts the iterative process of the OIRS alignment problem. Notably, the enhancement by OIRS is most significant in the corner of the room, i.e., region (0, 0), achieving an 8 times improvement when the number of OIRS units is 72. For the central region, i.e., region (4, 4), over 3 times enhancement is gained, which is smaller than region (0, 0) due to two factors. Firstly, the initial value of $e_{\min}$ in the corner area is the worst among all regions, making significant improvement easier. Secondly, the corner region is relatively closer to the OIRS, allowing the OIRS to have a more substantial impact. The results demonstrate the effectiveness of the proposed algorithm and highlight the substantial improvement in the evaluation indicator achieved by OIRS alignment.

The improvement in the evaluation indicator under different numbers of OIRS units for various regions is shown in Fig. 4. It is evident that the more OIRS units, the larger the

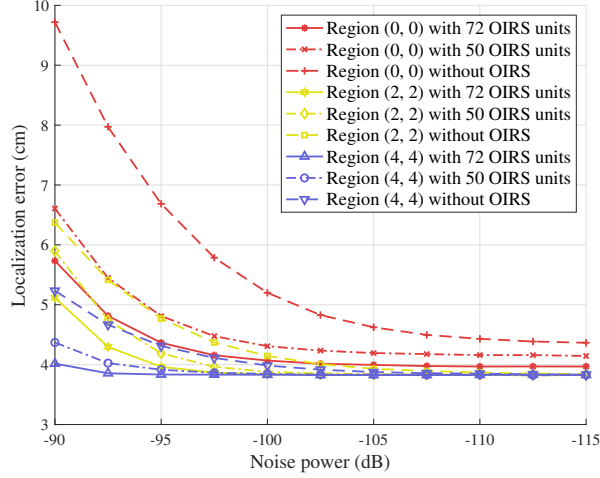


Fig. 5. The localization error versus the noise power under various scenarios.

enhancement in $e_{\min}$. The results indicate that the proposed scheme significantly improves the $e_{\min}$, with the improvement magnitude roughly linear with the number of OIRS units. Especially in the edge region, i.e., region (0, 0), the introduction of OIRS can improve the performance by an order of magnitude when the number of OIRS units reaches 80. Furthermore, the proposed algorithm is compared with a greedy alignment method [10] where the OIRS units prioritize serving the reference points associated with the current $e_{\min}$ for each iteration. It is proved that the proposed algorithm clearly outperforms the baseline approach.

Fig. 5 investigates the relationship between the localization error evaluated by the Monte Carlo algorithm and the noise power with varying numbers of OIRS units. The results indicate that lower noise power consistently results in better localization accuracy, which converges to an inherent error of 3.8 cm. Moreover, introducing OIRS to assist localization significantly enhances the positioning accuracy under the same noise power. For example, in region (0, 0), the localization error at the noise power of -95 dB is approximately 7 cm without OIRS. However, with OIRS assistance, this error is reduced to 4.5 cm, demonstrating a substantial improvement in the localization precision. Furthermore, when the noise power is -90 dB, the localization error in region (4, 4) with 72 OIRS units nearly achieves the inherent error, highlighting the benefit of OIRS in high noise power scenarios.

VI. CONCLUSION

This study explored an advanced VLP leveraging RSS fingerprint augmented by OIRS, which addressed the limitations in distinguishing near locations. Then, a two-stage positioning scheme was introduced to take advantage of OIRS. Besides, we defined the evaluation indicator to embody the positioning effectiveness, and then formulated the problem to maximize the indicator by optimizing the OIRS alignment. To solve the problem, an iterative algorithm based on SCA technique

was proposed. Moreover, the simulation reveals substantially enhancement in the evaluation indicator with OIRS. Besides, the impacts of the OIRS units number and noise power are also exhibited in the numerical results, indicating that the OIRS can potentially reduce the localization error and also maintain the positioning performance in high noise power condition. To sum up, these findings suggest that the incorporation of OIRS into VLP systems presents a promising approach for achieving higher accuracy in indoor positioning.

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